

What is Linear Algebra?

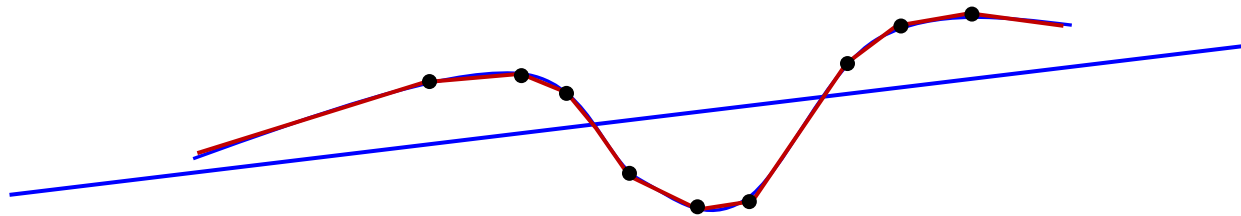
having the properties of lines

1-dimensional linear object: line

2-dimensional linear object: plane

linear systems
matrices
vectors

The study of mathematical objects and properties that are related to lines, planes and their generalization.



Linear Systems

Linear system

2 equations

2 variables

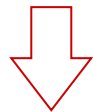
$$\begin{array}{rcl} x & + & y = 1 \\ 2x & + & 3y = 2 \end{array}$$

represented by 2 lines



solution: $x = 1, y = 0$ intersection of the 2 lines

algebraic
generalization



Linear system

5 equations

4 variables

$$\begin{array}{rcl} x & + & y & + & z & + & w & = & 1 \\ 2x & - & y & + & 3z & + & 5w & = & 2 \\ x & + & 2y & + & 7z & + & 0w & = & 5 \\ 0x & - & 6y & + & 2z & + & 9w & = & 0 \\ 5x & + & 2y & - & 4z & + & 7w & = & 8 \end{array}$$

How to find the solutions?

Heat Transfer Problem

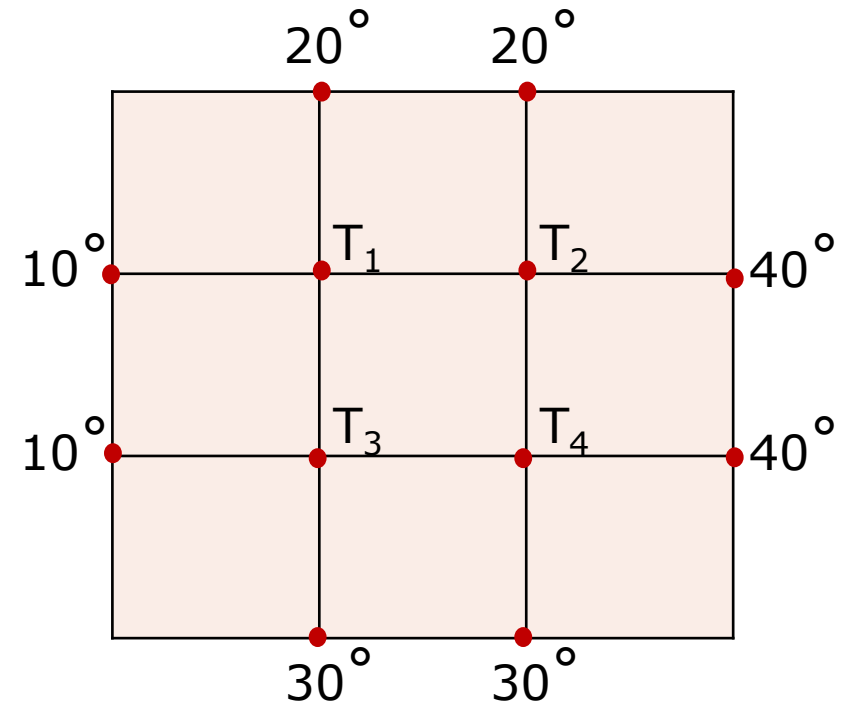
Steady-State temperature distribution of a thin plate

Temperature around boundary is known.

T_1, T_2, T_3, T_4 are temperatures at the 4 internal nodes.

Find T_1, T_2, T_3, T_4 .

Known fact: Temperature at a node is approximately equal to the average of the 4 nearest nodes (on the left, right, above, below).



Known fact: Temperature at a node is approximately equal to the average of the 4 nearest nodes (on the left, right, above, below).

Heat Transfer Problem

→ Set up a **Linear System** in variables T_1, T_2, T_3, T_4 .

Node 1: $T_1 = (10 + 20 + T_2 + T_3)/4$

$$4T_1 - T_2 - T_3 = 30$$

Node 2: $T_2 = (20 + 40 + T_4 + T_1)/4$

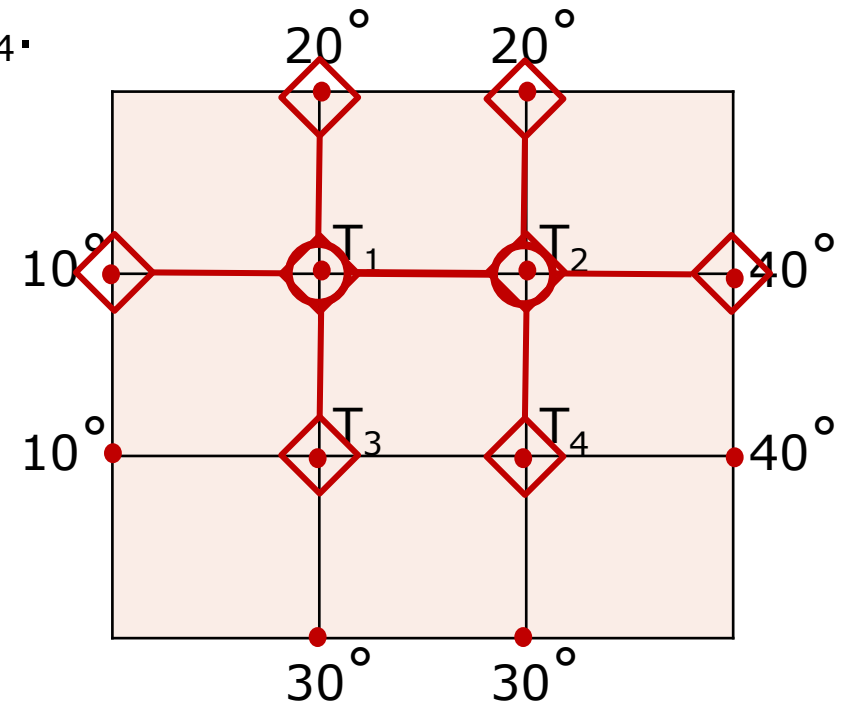
$$-T_1 + 4T_2 - T_4 = 60$$

Node 3: $T_3 = (10 + 30 + T_4 + T_1)/4$

$$-T_1 + 4T_3 - T_4 = 40$$

Node 4: $T_4 = (30 + 40 + T_2 + T_3)/4$

$$-T_2 - T_3 + 4T_4 = 70$$



Linear Systems – Different Expressions

x	$+$	y	$+$	z	$+$	w	$=$	1
$2x$	$-$	y	$+$	$3z$	$+$	$5w$	$=$	2
x	$+$	$2y$	$+$	$7z$	$+$	$0w$	$=$	5
$0x$	$-$	$6y$	$+$	$2z$	$+$	$9w$	$=$	0
$5x$	$+$	$2y$	$-$	$4z$	$+$	$7w$	$=$	8

standard form

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & -1 & 3 & 5 \\ 1 & 2 & 7 & 0 \\ 0 & -6 & 2 & 9 \\ 5 & 2 & -4 & 7 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 5 \\ 0 \\ 8 \end{pmatrix}$$

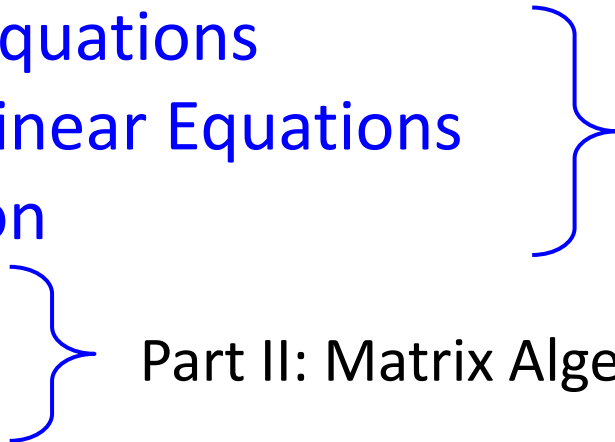
matrix equation form

$$x \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \\ 5 \end{pmatrix} + y \begin{pmatrix} 1 \\ -1 \\ 2 \\ -6 \\ 2 \end{pmatrix} + z \begin{pmatrix} 1 \\ 3 \\ 7 \\ 2 \\ -4 \end{pmatrix} + w \begin{pmatrix} 1 \\ 5 \\ 0 \\ 9 \\ 7 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 5 \\ 0 \\ 8 \end{pmatrix}$$

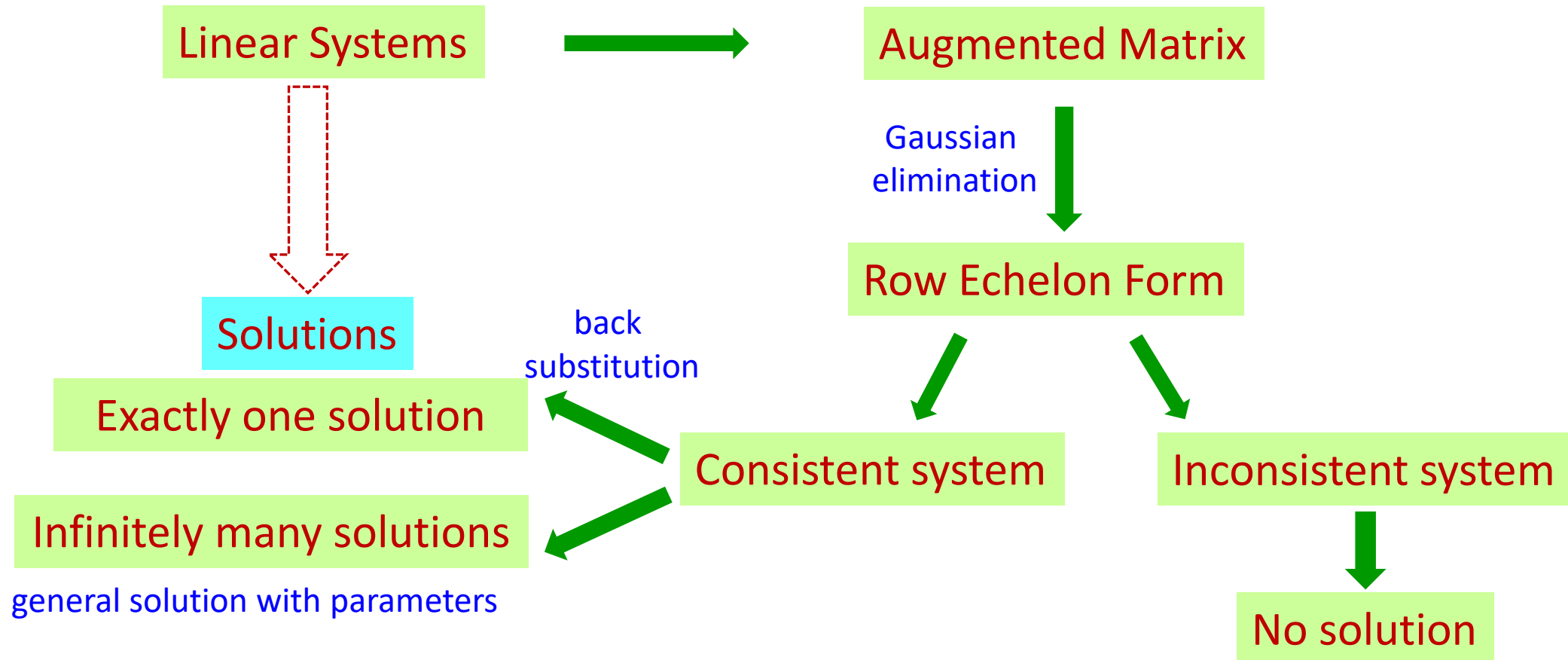
vector equation form

Topics for week 1

Chapter 1: Linear Systems and Matrix Algebra

- 1.1 Systems of Linear Equations
 - 1.2 Solving System of Linear Equations
 - 1.3 Gaussian Elimination
 - 1.4 Matrices
 - 1.5 Matrix inverses 1
- Part I: Linear Systems
- Part II: Matrix Algebra
- 

Part I: Linear Systems



Elementary Row Operations

Augmented Matrix

Gaussian
elimination



Row Echelon Form

1. Multiply a row by a nonzero constant.
2. Interchange two rows.
3. Add a multiple of one row to another row.

Elementary Row Operations

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 1 & 2 \\ -4 & -6 & 0 & 4 \\ 3 & 3 & 6 & -9 \end{pmatrix} \xrightarrow{\text{Interchange row 1 and 3}} \mathbf{B} = \begin{pmatrix} 3 & 3 & 6 & -9 \\ 2 & 3 & 0 & -2 \\ 0 & 0 & 1 & 2 \end{pmatrix}$$

Multiply row 3
by 1/3

$$\mathbf{C} = \begin{pmatrix} 0 & 0 & 1 & 2 \\ 2 & 3 & 0 & -2 \\ 1 & 1 & 2 & -3 \end{pmatrix}$$

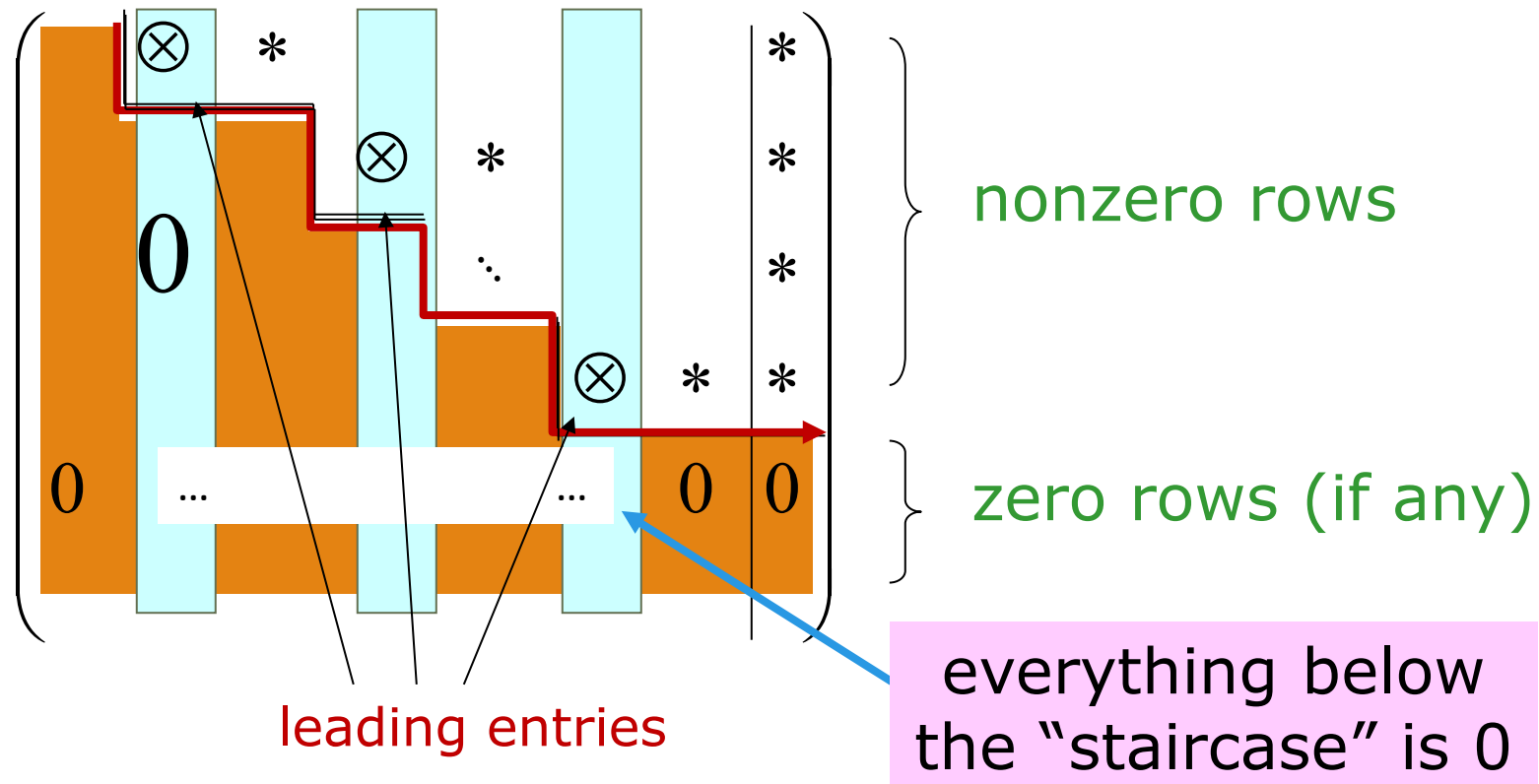
-2 x row 2 and
add to row 3

$$\mathbf{D} = \begin{pmatrix} 0 & 0 & 1 & 2 \\ 2 & 3 & 0 & -2 \\ -1 & -3 & 6 & -5 \end{pmatrix}$$

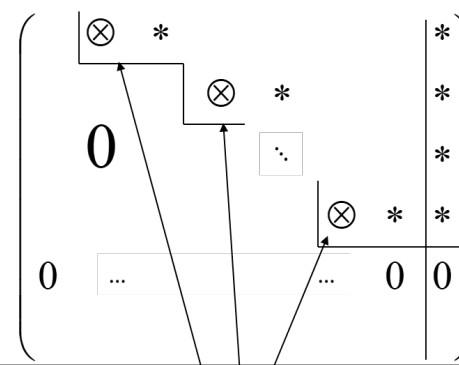
only row 3 is changed

Row Echelon Form

columns that contain leading entries called **pivot columns**



Row Echelon Form



leading entries

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

A

$$\begin{pmatrix} 0 & 0 & 0 & 1 & 3 \\ 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

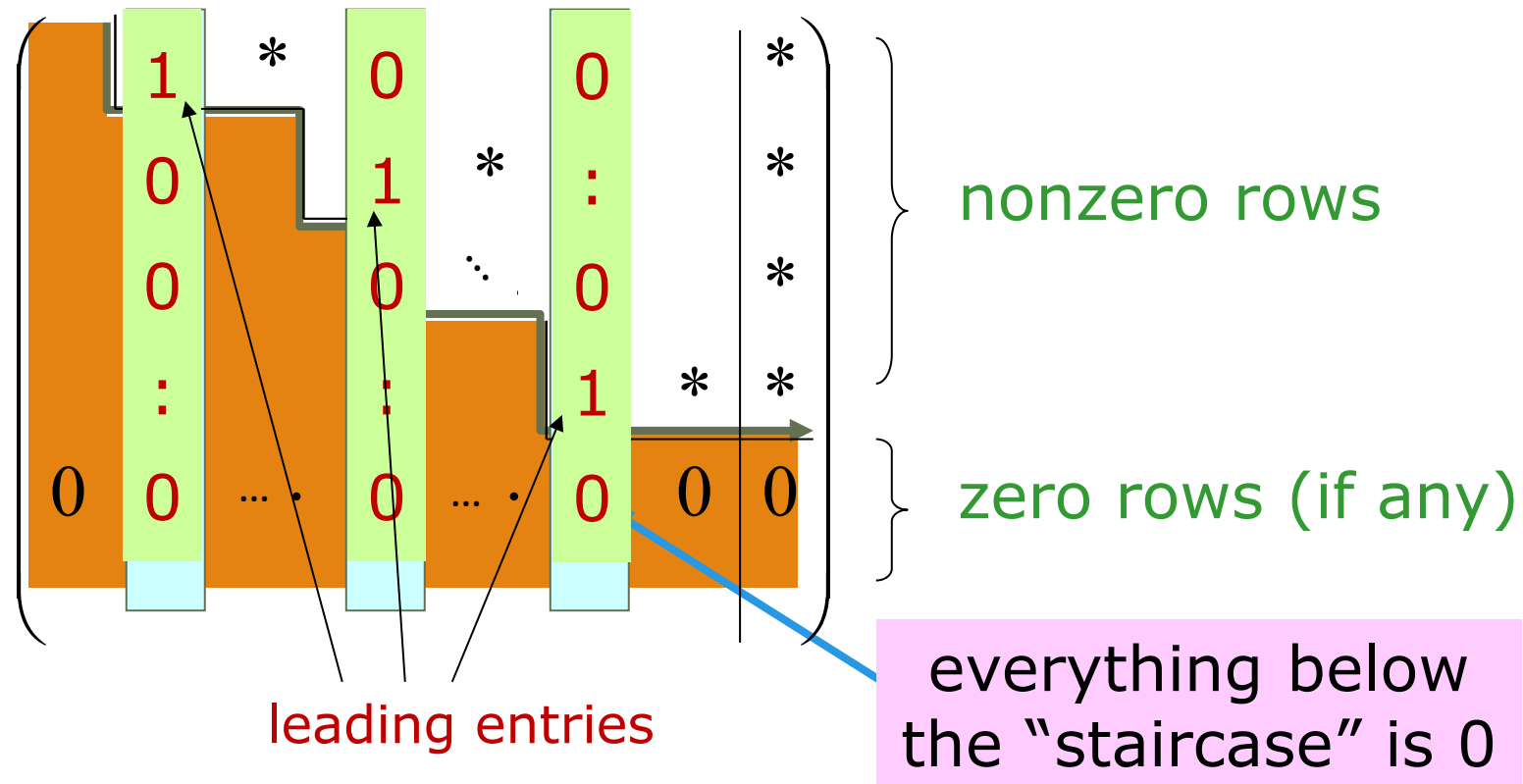
B

$$\begin{pmatrix} 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

C

Reduced Row Echelon Form

columns that contain leading entries called **pivot columns**



Reduced Row Echelon Form

A diagram of a matrix with leading entries highlighted in green. The matrix is partitioned into three columns and two rows. The first column has a leading entry of 1 in the first row. The second column has a leading entry of 1 in the second row. The third column has a leading entry of 1 in the second row. The matrix is partitioned into three columns and two rows. The first column has a leading entry of 1 in the first row. The second column has a leading entry of 1 in the second row. The third column has a leading entry of 1 in the second row. The matrix is partitioned into three columns and two rows. The first column has a leading entry of 1 in the first row. The second column has a leading entry of 1 in the second row. The third column has a leading entry of 1 in the second row.

$$\begin{pmatrix} 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Example 1

$$x_1 - 2x_2 - x_3 - x_4 = -4$$

$$3x_1 + x_2 + x_3 - 2x_4 = 11$$

$$x_1 + 12x_2 + 7x_3 + x_4 = 31$$

The augmented matrix is:

$$\left(\begin{array}{cccc|c} 1 & -2 & -1 & -1 & -4 \\ 3 & 1 & 1 & -2 & 11 \\ 1 & 12 & 7 & 1 & 31 \end{array} \right)$$

Example 1 (Gaussian Elimination)

$$\left(\begin{array}{cccc|c} 1 & -2 & -1 & -1 & -4 \\ 3 & 1 & 1 & -2 & 11 \\ 1 & 12 & 7 & 1 & 31 \end{array}\right) \xrightarrow[R_3 - R_1]{R_2 - 3R_1} \left(\begin{array}{cccc|c} 1 & -2 & -1 & -1 & -4 \\ 0 & 7 & 4 & 1 & 23 \\ 0 & 14 & 8 & 2 & 35 \end{array}\right)$$

$$\xrightarrow{R_3 - 2R_2} \left(\begin{array}{cccc|c} 1 & -2 & -1 & -1 & -4 \\ 0 & 7 & 4 & 1 & 23 \\ 0 & 0 & 0 & 0 & -11 \end{array}\right)$$

$0x_1 + 0x_2 + 0x_3 + 0x_4 = -11$

The linear system has no solution

We say the system is **inconsistent**

Example 2

Homogeneous system

$$-2x_3 + 7x_5 + 12x_6 = 0$$

$$2x_1 + 4x_2 - 10x_3 + 6x_4 + 12x_5 + 28x_6 = 0$$

$$2x_1 + 4x_2 - 5x_3 + 6x_4 - 5x_5 - x_6 = 0$$

The augmented matrix

$$\left(\begin{array}{cccccc|c} 0 & 0 & -2 & 0 & 7 & 12 & 0 \\ 2 & 4 & -10 & 6 & 12 & 28 & 0 \\ 2 & 4 & -5 & 6 & -5 & -1 & 0 \end{array} \right)$$

Example 2 (Free parameters)

A row echelon form is

$$\left(\begin{array}{cccc|cc} 1 & 2 & -5 & 3 & 6 & 14 & 0 \\ 0 & 0 & 1 & 0 & -7/2 & -6 & 0 \\ 0 & 0 & 0 & 0 & 1 & 2 & 0 \end{array} \right)$$

The 3 variables corresponding to the non-pivot columns

$$x_1 + 2x_2 - 5x_3 + 3x_4 + 6x_5 + 14x_6 = 0$$

$$x_3 - \frac{7}{2}x_5 - 6x_6 = 0$$

$$x_5 + 2x_6 = 0$$

3 equations, 6 variables

need to set 6 - 3 variables as free parameters

Which 3 should we set as parameters?

$$\left(\begin{array}{ccc|ccc|c} 1 & 2 & -5 & 3 & 6 & 14 & 0 \\ 0 & 0 & 1 & 0 & -7/2 & -6 & 0 \\ 0 & 0 & 0 & 0 & 1 & 2 & 0 \end{array}\right)$$

Example 2 (Back substitution)

$$x_1 + 2x_2 - 5x_3 + 3x_4 + 6x_5 + 14x_6 = 0 \quad \leftarrow$$

$$x_3 - \frac{7}{2}x_5 - 6x_6 = 0 \quad \leftarrow$$

$$x_5 + 2x_6 = 0 \quad \leftarrow$$

Let $x_6 = t$ $x_2 = r$ and $x_4 = s$

Then $x_5 = -2x_6 = -2t$

Then $x_3 = \frac{7}{2}x_5 + 6x_6 = -t$

$$x_1 = -2x_2 + 5x_3 - 3x_4 - 6x_5 - 14x_6$$

$$= -2r + 5(-t) - 3s - 6(-2t) - 14t = -2r - 3s - 7t$$

Example 2 (General Solution)


$$x_1 = -2r - 3s - 7t$$

$$x_2 = r$$

$$x_3 = -t$$

$$x_4 = s$$

$$x_5 = -2x_6 = -2t$$

$$x_6 = t$$

r, s, t any real numbers

The linear system has infinitely many solutions

$$\begin{aligned}
 & -2x_3 + 7x_5 + 12x_6 = 0 \\
 2x_1 + 4x_2 - 10x_3 + 6x_4 + 12x_5 + 28x_6 &= 0 \\
 2x_1 + 4x_2 - 5x_3 + 6x_4 - 5x_5 - x_6 &= 0
 \end{aligned}$$

Homogeneous Systems

$$x_1 = -2r - 3s - 7t$$

$$x_2 = r$$

$$x_3 = -t$$

$$x_4 = s$$

$$x_5 = -2x_6 = -2t$$

$$x_6 = t$$

Linear Systems

Homogeneous system

(always consistent)

Non-homogeneous system

(may be inconsistent)

No parameter
in solution

Only trivial solution

At least 1 parameter
in general solution

Trivial +
non-trivial solutions

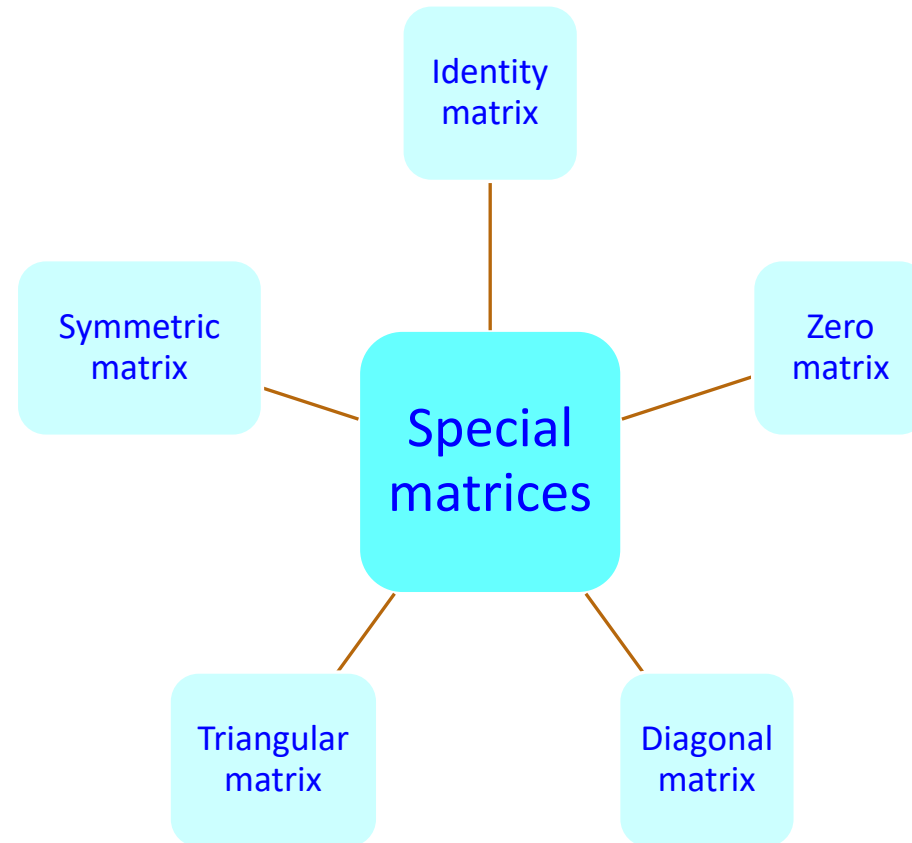
Part II: Matrix Algebra

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

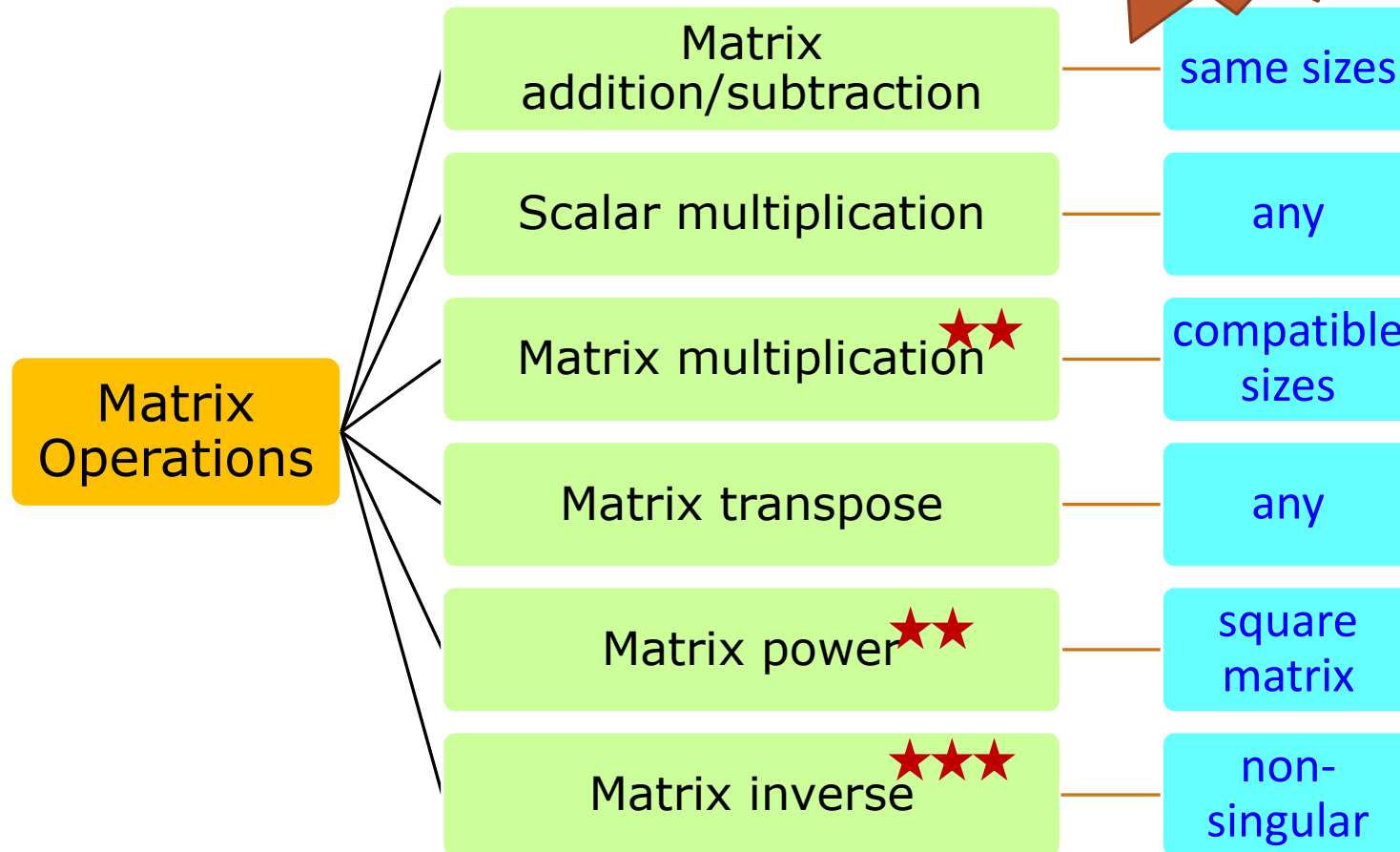
Terms associated to matrix

- **rows** (m)
- **columns** (n)
- **size** (m x n)
- **entries** (a_{ij})



Part II: Matrix Algebra

Conditions
for
operations



Matrix Multiplication

$$\begin{array}{c}
 \mathbf{a}_1 \\
 \mathbf{a}_2 \\
 \mathbf{a}_3
 \end{array}
 \begin{array}{c}
 \mathbf{A} \\
 \left(\begin{array}{cc}
 1 & 1 \\
 2 & 3 \\
 -1 & -2
 \end{array} \right)
 \end{array}
 \begin{array}{c}
 \mathbf{B} \\
 \left(\begin{array}{ccc}
 1 & 2 & 3 \\
 4 & 5 & 6
 \end{array} \right)
 \end{array}
 \begin{array}{c}
 \mathbf{b}_1 \quad \mathbf{b}_2 \quad \mathbf{b}_3
 \end{array}
 =
 \begin{array}{c}
 \mathbf{AB} \\
 \left(\begin{array}{ccc}
 5 & 7 & 9 \\
 14 & 19 & 24 \\
 -9 & -12 & -15
 \end{array} \right)
 \end{array}
 \begin{array}{c}
 \mathbf{a}_1 \mathbf{b}_1 \quad \mathbf{a}_1 \mathbf{b}_2 \quad \mathbf{a}_1 \mathbf{b}_3 \\
 \mathbf{a}_2 \mathbf{b}_1 \quad \mathbf{a}_2 \mathbf{b}_2 \quad \mathbf{a}_2 \mathbf{b}_3 \\
 \mathbf{a}_3 \mathbf{b}_1 \quad \mathbf{a}_3 \mathbf{b}_2 \quad \mathbf{a}_3 \mathbf{b}_3
 \end{array}$$

Example 3 (Matrix)

Three products A, B, C

Production cost per item

	A	B	C
Raw materials	0.10	0.30	0.15
Labor	0.30	0.40	0.25
Overhead & Misc.	0.10	0.20	0.13

$$\begin{pmatrix} 0.10 & 0.30 & 0.15 \\ 0.30 & 0.40 & 0.25 \\ 0.10 & 0.20 & 0.13 \end{pmatrix}$$

Production cost matrix

Easier to manipulate the data

Example 3 (Matrix Multiplication)

Production cost per item

	A	B	C
Raw materials	0.10	0.30	0.15
Labor	0.30	0.40	0.25
Overhead & Misc.	0.10	0.20	0.13

produce: 3000 of A
2000 of B
5000 of C

$$\begin{pmatrix} 0.10 & 0.30 & 0.15 \\ 0.30 & 0.40 & 0.25 \\ 0.10 & 0.20 & 0.13 \end{pmatrix} \begin{pmatrix} 3000 \\ 2000 \\ 5000 \end{pmatrix} = \begin{pmatrix} 1650 \\ 2950 \\ 1350 \end{pmatrix}$$

total raw material cost
total labor cost
total overhead cost

Example 3 (Matrix Inverse)

Production cost per item

	A	B	C
Raw materials	0.10	0.30	0.15
Labor	0.30	0.40	0.25
Overhead & Misc.	0.10	0.20	0.13

Given budget:

2000 for raw material

2500 for labor

1500 for overhead cost

How many of A, B, C can be produce?

$$\begin{pmatrix} 0.10 & 0.30 & 0.15 \\ 0.30 & 0.40 & 0.25 \\ 0.10 & 0.20 & 0.13 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 2000 \\ 2500 \\ 1500 \end{pmatrix} \rightarrow \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0.10 & 0.30 & 0.15 \\ 0.30 & 0.40 & 0.25 \\ 0.10 & 0.20 & 0.13 \end{pmatrix}^{-1} \begin{pmatrix} 2000 \\ 2500 \\ 1500 \end{pmatrix}$$

Example 4 (A simple model)

In a certain town, 30% of married women get divorced each year and 20% of women get married each year.

There are 8000 married women and 2000 single women.

Assuming that the total population of women remains constant, how many married women and how many single women will there be after n years?

M_n = married women after n years

$$M_n = 0.7M_{n-1} + 0.2S_{n-1}$$

S_n = single women after n years

$$S_n = 0.3M_{n-1} + 0.8S_{n-1}$$

$$\begin{pmatrix} M_n \\ S_n \end{pmatrix} = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix} \begin{pmatrix} M_{n-1} \\ S_{n-1} \end{pmatrix}$$

Example 4 (A simple model)

In a certain town, 30% of married women get divorced each year and 20% of women get married each year.

There are 8000 married women and 2000 single women.

Assuming that the total population of women remains constant, how many married women and how many single women will there be after n years?

After 1 year

$$\begin{pmatrix} M_1 \\ S_1 \end{pmatrix} = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix} \begin{pmatrix} 8000 \\ 2000 \end{pmatrix} = \begin{pmatrix} 6000 \\ 4000 \end{pmatrix}$$

After 2 year

$$\begin{pmatrix} M_2 \\ S_2 \end{pmatrix} = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix} \begin{pmatrix} 6000 \\ 4000 \end{pmatrix} = \begin{pmatrix} 5000 \\ 5000 \end{pmatrix} = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}^2 \begin{pmatrix} 8000 \\ 2000 \end{pmatrix}$$

Example 4 (Matrix Power)

In a certain town, 30% of married women get divorced each year and 20% of women get married each year.

There are 8000 married women and 2000 single women.

Assuming that the total population of women remains constant, how many married women and how many single women will there be after n years?

After n year

n times

$$\begin{pmatrix} M_n \\ S_n \end{pmatrix} = \overbrace{\begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix} \cdots \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix} \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}}^{n \text{ times}} \begin{pmatrix} 8000 \\ 2000 \end{pmatrix}$$
$$= \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}^n \begin{pmatrix} 8000 \\ 2000 \end{pmatrix}$$